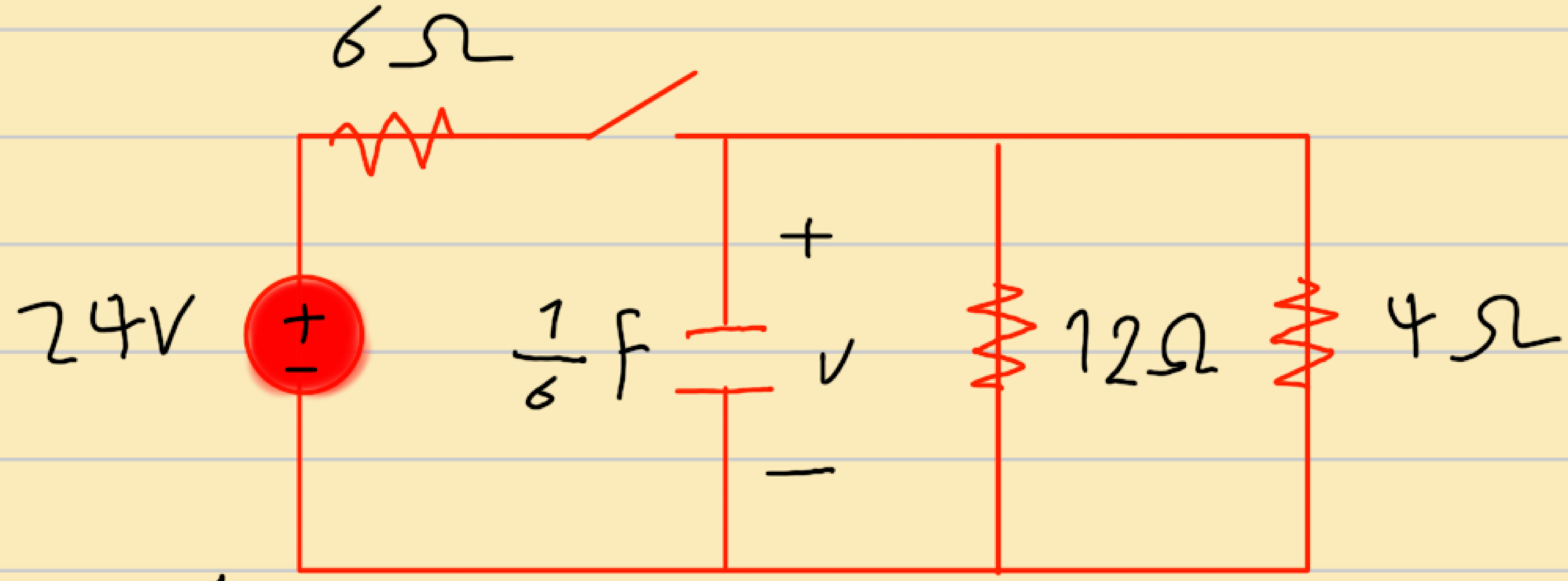
FINAL

RC circuit:-

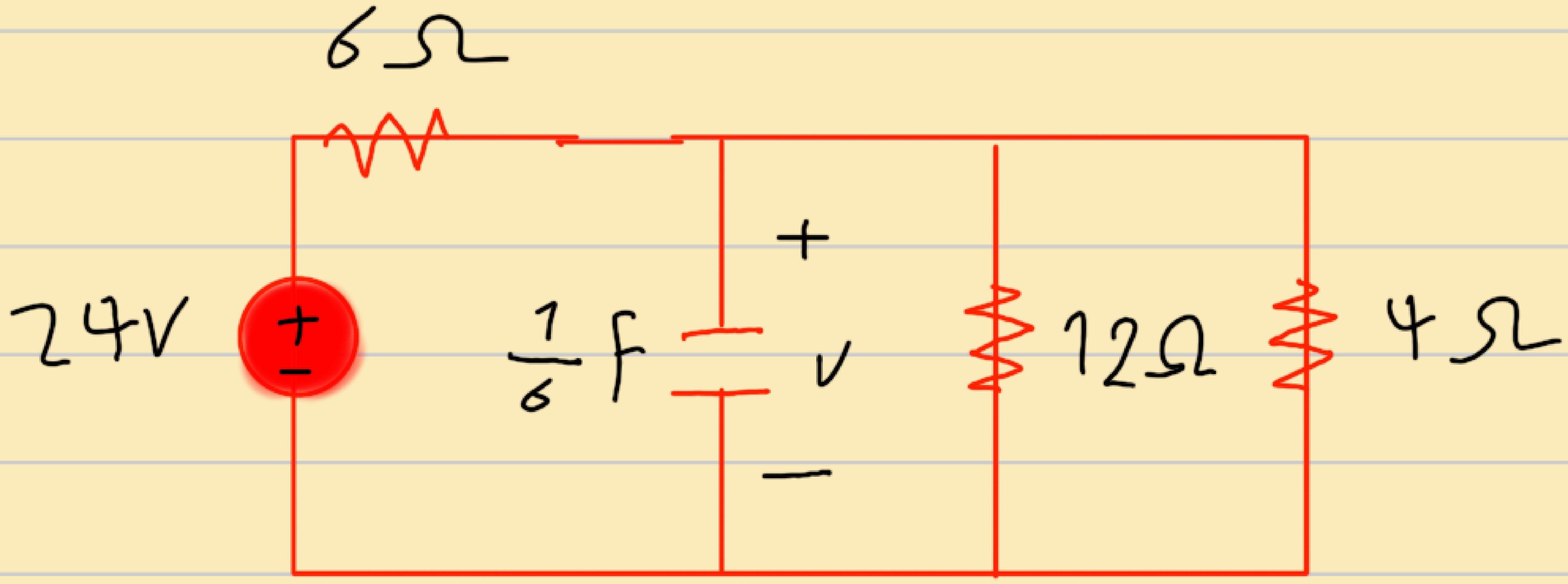
In the circuit in the figure, the switch has been closed for a long time. It is opened at $t=0$.

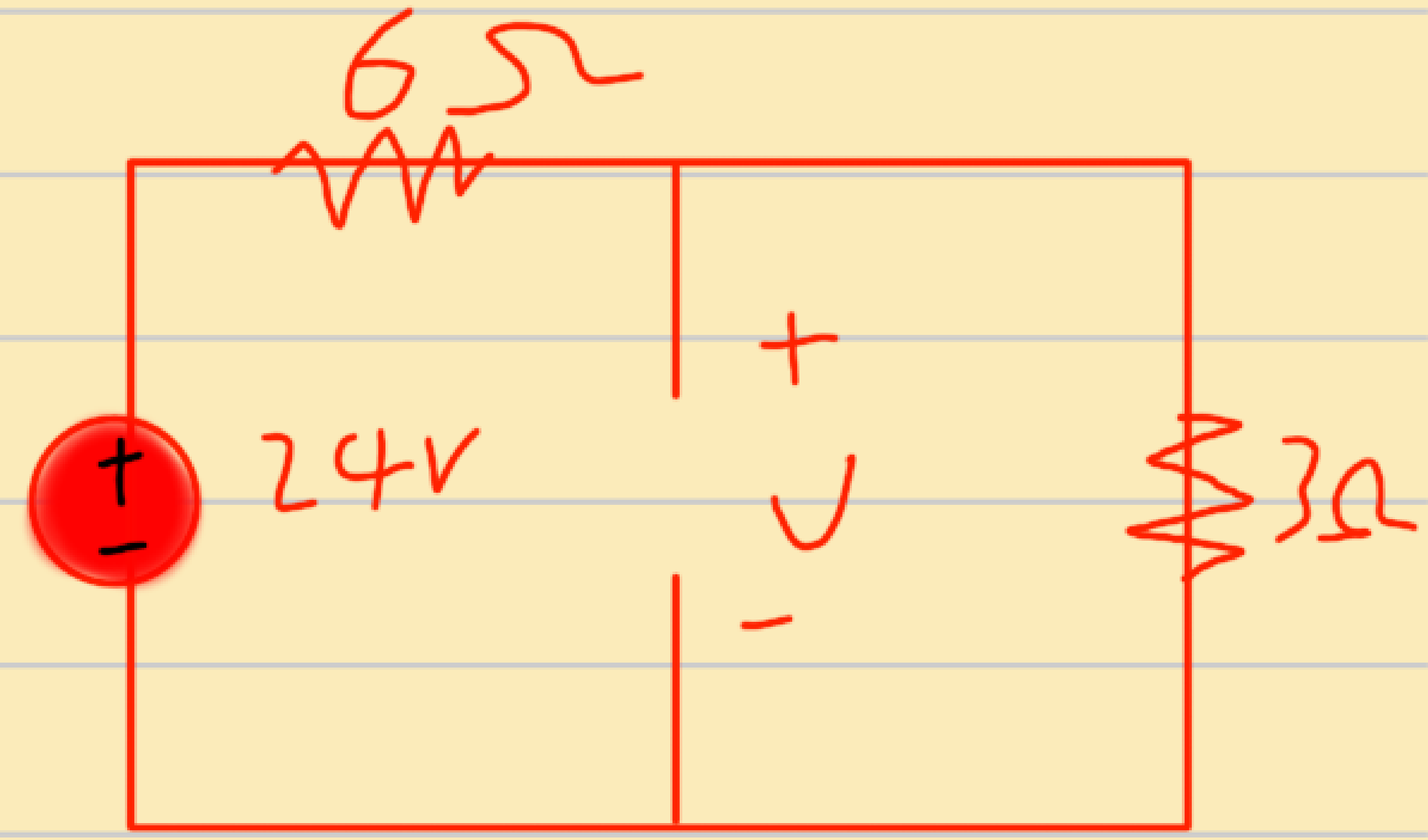


- Find the energy stored in the capacitor at the initial time.

- Find the capacitor voltage at time t.

First :-





first find Vc :-

$$V_c = \frac{3}{6+3} 24 = 8V \qquad V_o = 8V$$

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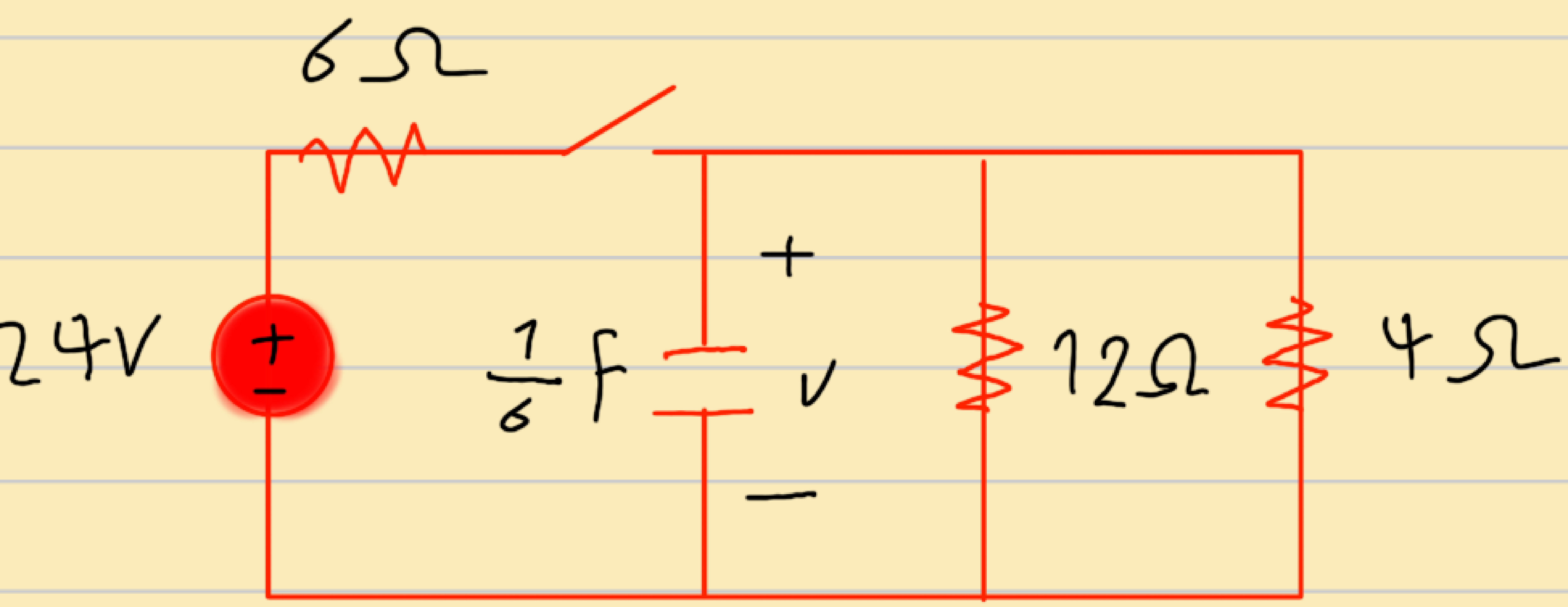
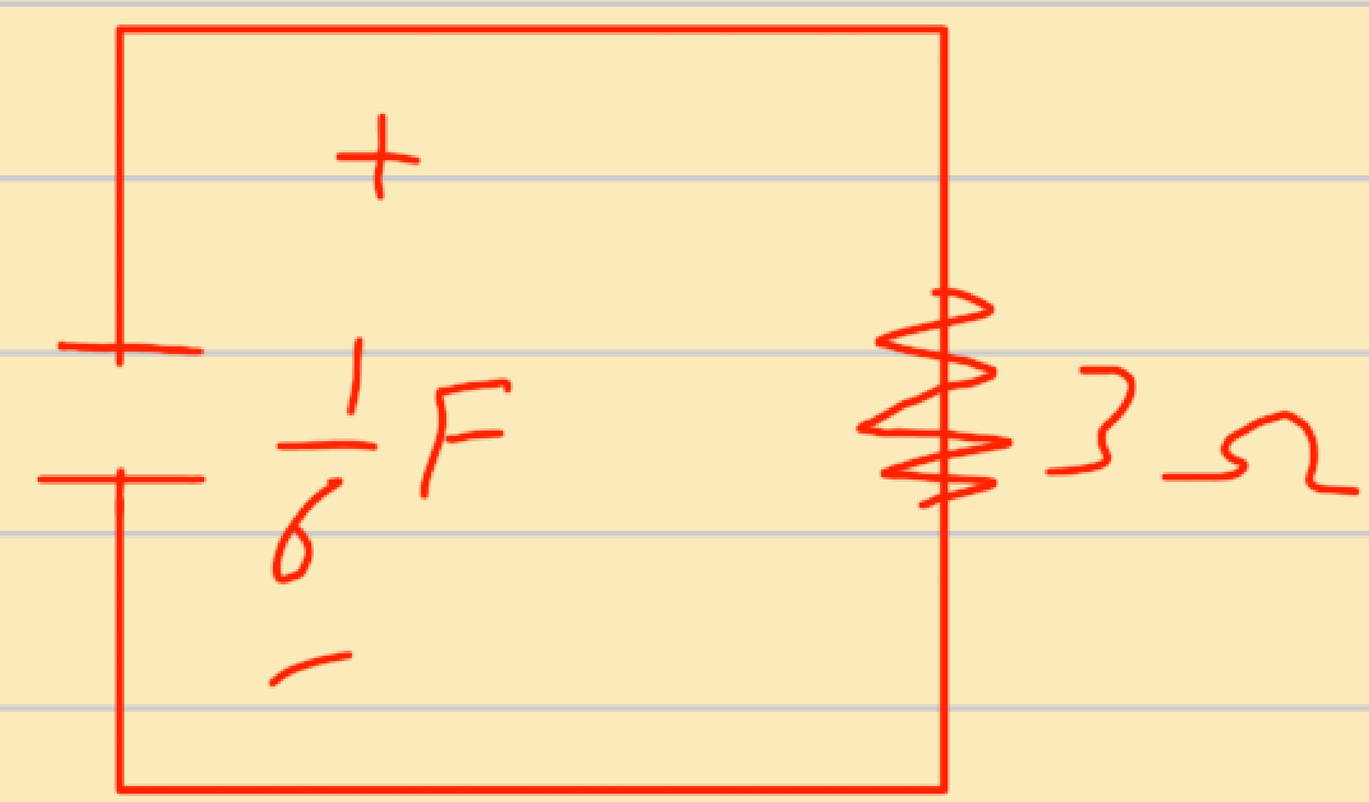
second :-





First find τ :-

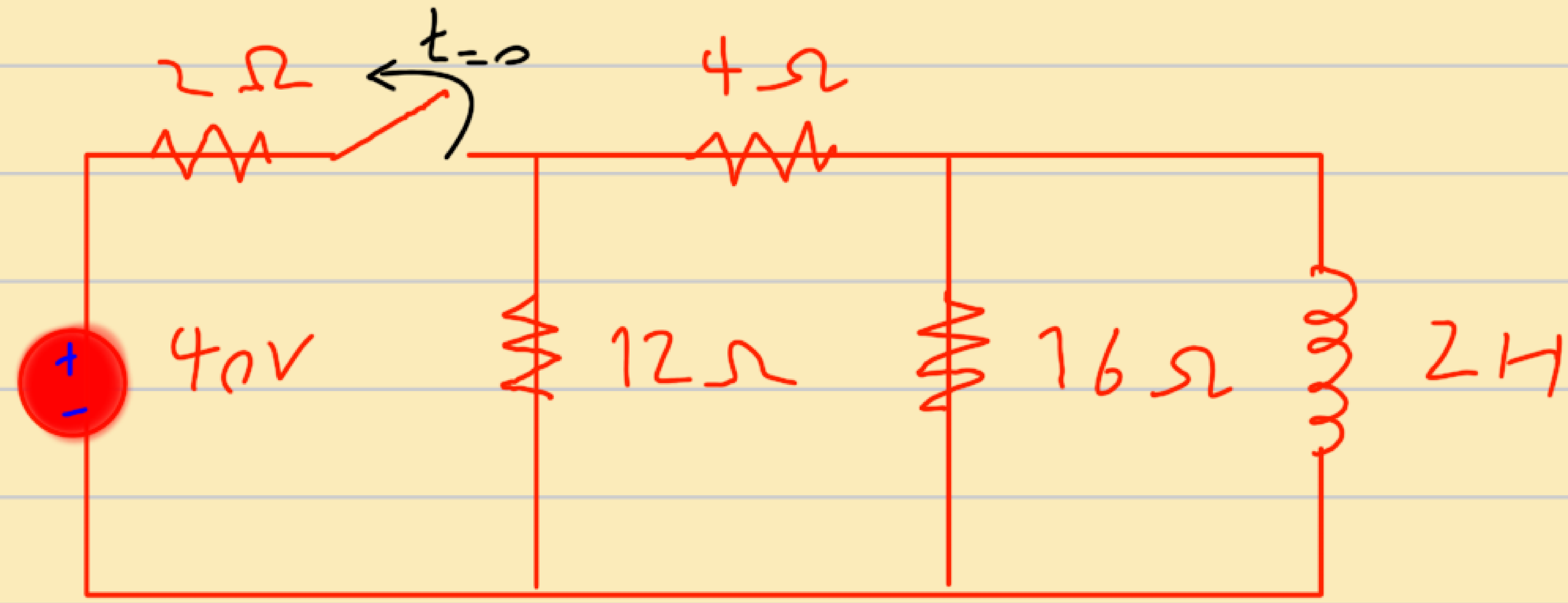
$\tau = RC = 3 \times \frac{1}{6} = 0.5 s$

second find $V_c(t)$:-

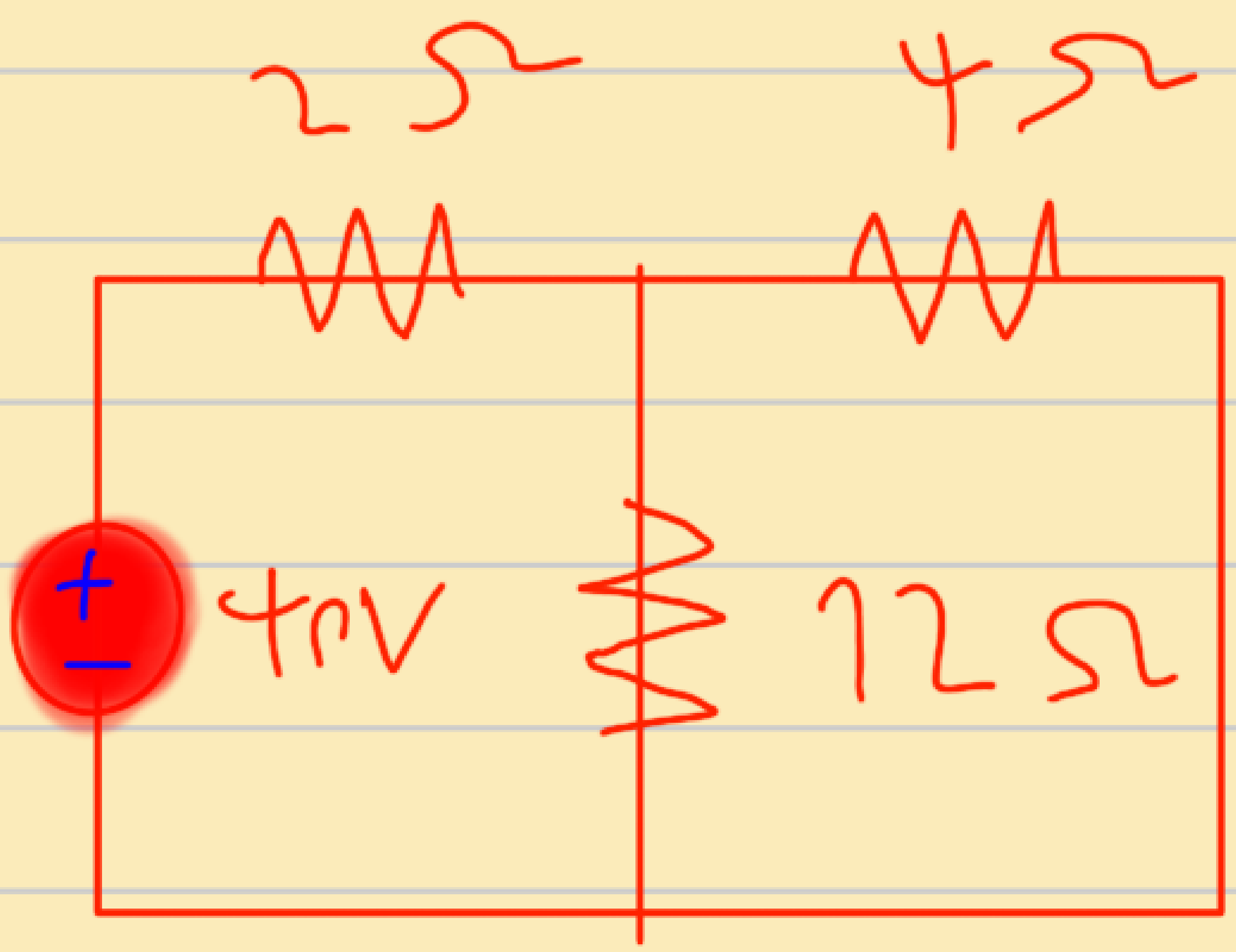
$v_c(t) = V_o e^{-t/RC} = 8 e^{-t/0.5} = 8 e^{-2t} V$

RL circuit:-

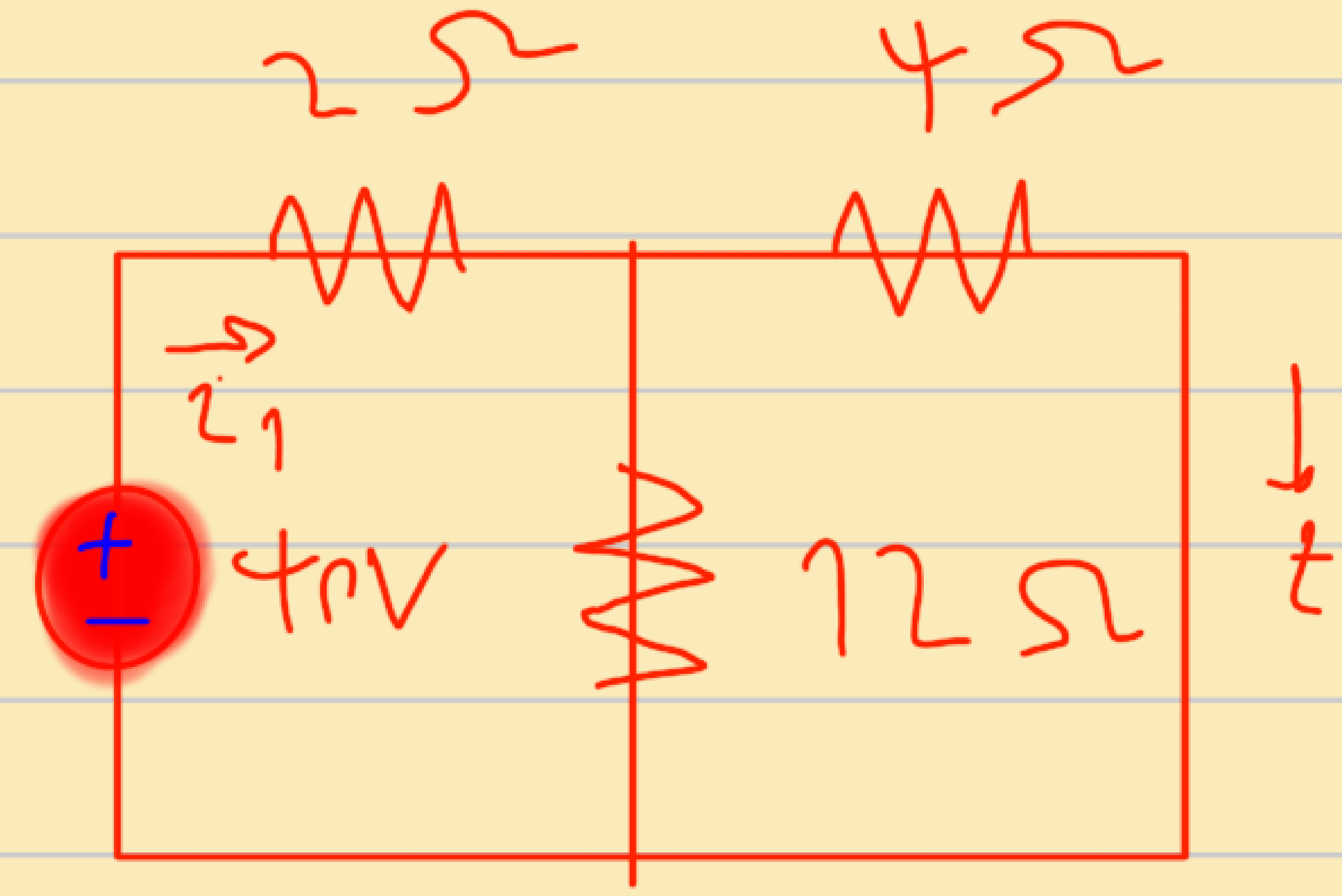
The switch remains closed for a sufficient time. At time $t=0$ the switch opens. Calculate i_t for $t>0$.



When the switch is closed the coil will act as a short circuit. therefore the 16 Ω resistor will be short circuited.



first step find Req:-



$$R_{eq} = \frac{2 \times 12}{2+12} = 1.71\,\Omega$$

$$1.71 + 4 = 5.71\,\Omega$$

second step find i_1 wich flwing in Req:-

$$i_1 = \frac{V}{R_{eq}} = \frac{40}{5.71} = 7.005\,A$$

therd step find $i(t)$:-

- from the current Divider circuit Rule.

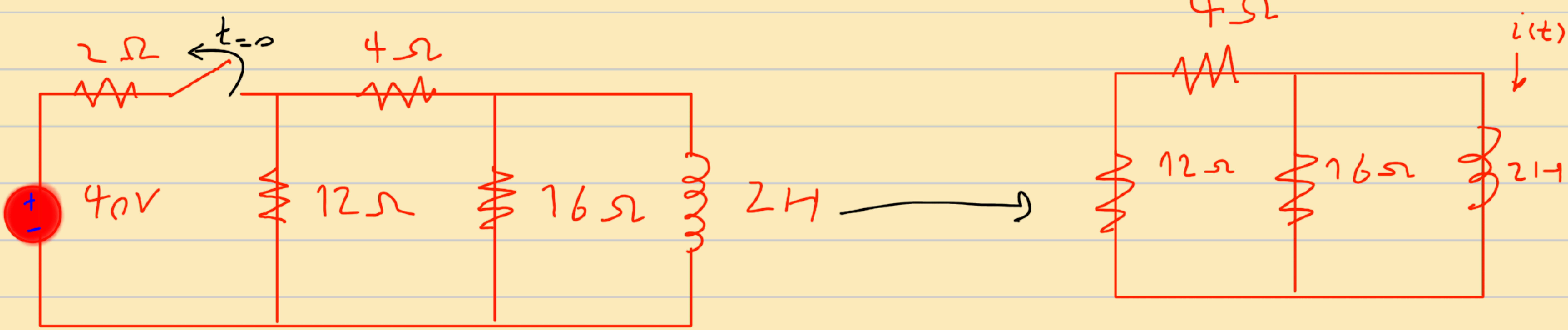
! $i(t)$ wich is flowing in short circuit !

$$i(t) = \frac{12}{12+4}\,7.005 = 5.25\,A$$

• Since the current in the inductor cannot change instantly;

$$i(0) = i(0^-) = 3.25\ A$$

After the switch is opened the voltage source will be deactivated.



First step find R_{eq} :-

$$R_{eq} = 12 + 4 = 16$$

$$\frac{16 \times 16}{16 + 16} = 8\ \Omega$$

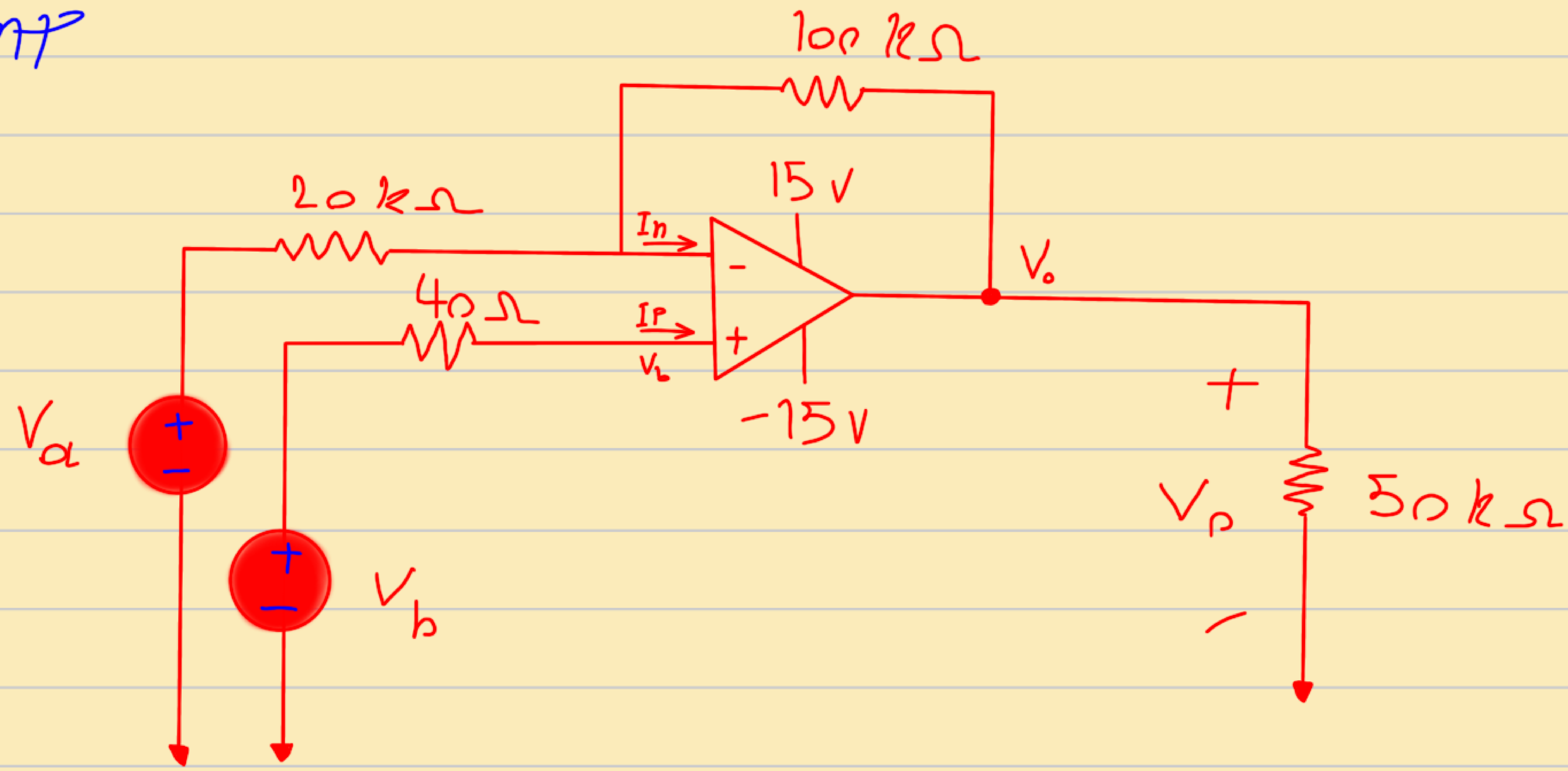
seconde step find τ:-

$$\tau = \frac{L}{R_{eq}} = \frac{2}{8} = 0.25\,s$$

Last step Find $i(t)$:-

$$i(t) = i_o e^{-t/RL} \qquad i(t) = 5.25\,e^{-t/0.25}\,A = 5.25\,e^{-4t}\,A$$

OPamp



a) Given that $V_a = 4V$, $V_b = 0V$ then Find V_o

b) Given that $V_a = 2V$, $V_b = 0V$ then find V_o

c) Given that $V_a = 2V$, $V_b = 1V$ then find V_o

d) Given that $V_a = 1V$, $V_b = 2V$ then find V_o

first find the equation:-

$$\frac{V_n - V_a}{20} + \frac{V_n - V_o}{100} = 0$$

$I_n = I_P = 0 \longrightarrow V_n = V_p = V_b$

$$\frac{V_b - V_a}{20} + \frac{V_b - V_o}{100} = 0 \qquad 5V_b - 5V_a + 1V_b = V_o$$

$$6V_b - 5V_a = V_o$$

a) $V_a = 4V$ $V_b = 0V$

$6(0) - 5(4) = -20 \longrightarrow a = -20$

b) $V_a = 2V$ $V_b = 0V$

$6(0) - 5(2) = -10 \longrightarrow b = -10$

c) $V_a = 2V$ $V_b = 1V$

$6(1) - 5(2) = -4 \longrightarrow c = -4$

d) $V_a = 1V$ $V_b = 2V$

$6(2) - 5(1) = 7 \longrightarrow d = 7$